

Your Full Name

✉ you@university.edu | ☎ +1 (555) 000-0000 | yoursite.com | github.com/yourhandle | City, State

Research interests: Numerical Analysis · Scientific Computing · Stochastic Processes · Optimization · Mathematical Modeling

EDUCATION

Ph.D. in Applied Mathematics *University of Somewhere*

Expected May 2026

Advisor: Prof. Jane Doe | GPA: 3.95/4.0

- Dissertation: "Adaptive Multilevel Methods for Stochastic PDEs on Complex Domains"
- Coursework: Numerical PDEs Functional Analysis Stochastic Calculus Optimization ML Theory

B.S. Mathematics (Honors), Minor Computer Science *State University*

May 2020

Summa Cum Laude | GPA: 3.98/4.0

- Senior thesis: "Finite Element Approximations to the Navier–Stokes Equations"

RESEARCH EXPERIENCE

Graduate Research Assistant *Computational Science Lab, University of Somewhere*

Aug 2020 – Present

Advisor: Prof. Jane Doe

- Developed adaptive multilevel Monte Carlo (MLMC) for elliptic SPDEs achieving $\mathcal{O}(h^2\epsilon^{-2})$ complexity — 3× speedup over standard MLMC on benchmark problems.
- Implemented parallel C++/MPI solver on unstructured meshes; near-linear weak scaling to 512 cores on university HPC cluster.
- Applied sparse-grid collocation for UQ in subsurface flow models across 20-dimensional parameter spaces.

Research Intern — Mathematics of Machine Learning *National Lab or Industry Partner*

Jun – Aug 2022

Mentor: Dr. John Smith

- Proved sharp $\mathcal{O}(k^{-1/2})$ SGD convergence under (L, σ) -smooth assumptions; derived matching lower bounds.
- Benchmarked quasi-Newton methods for large-scale logistic regression, outperforming L-BFGS by 18% on standard NLP datasets.

REU Participant — Dynamical Systems *Another University*

May – Jul 2019

Mentor: Prof. Alice Brown

- Computer-assisted proof of Hopf bifurcation in a 3-species Lotka–Volterra model using interval arithmetic.

PUBLICATIONS & PREPRINTS

Your Name, Jane Doe. "Adaptive Multilevel Monte Carlo for Stochastic Elliptic Equations." **SIAM Journal on Scientific Computing**, 2024. [\[link\]](#)

Your Name, John Smith. "Sharp Convergence Rates for Nonconvex SGD with Variance Reduction." **arXiv preprint arXiv:2401.00000**, 2023. [\[link\]](#)

Your Name, Alice Brown. "Computer-Assisted Proof of Hopf Bifurcation in a Predator–Prey System." **Journal of Mathematical Biology**, 2020.

TALKS & PRESENTATIONS

SIAM Annual Meeting	"Adaptive MLMC for SPDEs" (contributed talk)	2024
Applied Math Seminar	University of Somewhere (invited)	2023
ICIAM	"Nonconvex SGD convergence rates" (poster)	2023
AMS Spring Sectional	"Hopf bifurcations via interval arithmetic" (poster)	2020

TEACHING EXPERIENCE

Teaching Assistant — Numerical Analysis (MATH 5XX) *University of Somewhere*

Lead sections, write & grade problem sets, hold office hours for 45 students

Fall 2021, Spring 2023

Instructor of Record — Calculus II (MATH 1XX) *University of Somewhere*

Sole instructor; course evaluations 4.7/5.0

Summer 2022

TECHNICAL SKILLS

Languages: Python (NumPy, SciPy, JAX), C/C++, MATLAB, Julia, R, Bash

HPC & Parallel: MPI, OpenMP, SLURM, GPU computing (CUDA basics)

Math Software: FEniCS, PETSc, deal.II, Firedrake, LAPACK/BLAS

ML / Data: PyTorch, scikit-learn, pandas, Jupyter

Tools: Git, $\LaTeX$, Linux, Docker, CMake

Methods: FEM, FDM, Spectral Methods, MLMC, Optimization, Bayesian Inference

HONORS, FELLOWSHIPS & AWARDS

NSF Graduate Research Fellowship

2021–2026

SIAM Student Paper Prize (honorable mention)

2023

Dean's Doctoral Excellence Award, University of Somewhere

2022

Phi Beta Kappa, State University

2020

Barry Goldwater Scholarship

2019

SERVICE & PROFESSIONAL ACTIVITIES

Referee: *SIAM J. Numer. Anal.*, *Math. Comp.*, *J. Comput. Phys.* · Organizer: Graduate Applied Math Seminar, 2022–2024 ·

Member: SIAM, AMS

Project Portfolio

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SELECTED RESEARCH PROJECTS

Adaptive Multilevel Monte Carlo for Stochastic PDEs

2021–Present

Ph.D. dissertation project, University of Somewhere

[Numerical Analysis] [UQ] [C++/MPI] [SIAM J. Sci. Comput.]

Problem. UQ for subsurface flow requires statistics of solutions to elliptic PDEs $-\nabla \cdot (a(\mathbf{x}, \omega) \nabla u) = f$ with log-normal random coefficient a . Standard Monte Carlo costs $\mathcal{O}(\varepsilon^{-2} h^{-d})$ for RMSE ε .

Contribution. I designed an *adaptive* multilevel estimator that selects mesh levels $\{h_\ell\}$ and sample counts $\{N_\ell\}$ online by minimising cost subject to a variance budget, using a Gaussian process surrogate to predict level contributions.

- Proved optimal complexity $\mathcal{O}(\varepsilon^{-2} |\log \varepsilon|^2)$ without *a priori* knowledge of coefficient regularity.
- Implemented in C++17/MPI with FEniCS and PETSc; near-linear weak scaling to 512 MPI tasks on Stampede2.
- $3\times$ wall-clock speedup over standard MLMC on a 3-D Darcy flow benchmark with Matérn-3/2 coefficient.

Key result: Paper accepted, *SIAM J. Sci. Comput.* 2024. Code: github.com/yourhandle/adaptive-mlmc.

Sharp Convergence Analysis of Nonconvex SGD

Summer 2022

Research internship, National Lab

[Optimization Theory] [ML Theory] [Python/JAX] [arXiv preprint]

Problem. Convergence guarantees for SGD on nonconvex objectives remain loose; prior bounds require bounded gradients or structural assumptions rarely satisfied in practice.

Contribution. I established a sharp $\mathcal{O}(k^{-1/2})$ rate for $\mathbb{E}[\|\nabla f(x_k)\|^2]$ under (L, σ) -smoothness — strictly weaker than standard L -smoothness — via a Lyapunov function accounting for batch-size-dependent variance.

- Derived tight lower bounds confirming $\mathcal{O}(k^{-1/2})$ is unimprovable without additional curvature information.
- Validated on CIFAR-10/100; empirical convergence matched predicted rates across 6 architectures.
- Compared AdaGrad, Adam, SGD-M; identified regimes where adaptive methods provably dominate.

Key result: Preprint arXiv:2401.00000; under revision for NeurIPS 2024.

Sparse-Grid Collocation for High-Dimensional Bayesian Inversion

Spring 2023

Independent study / course project

[Bayesian Inference] [Sparse Grids] [Python] [SciPy/Chaospy]

Problem. Bayesian inversion of distributed parameter fields leads to posteriors on high-dimensional ($d = 50$ – 200) spaces; MCMC is prohibitive when each forward solve costs ~ 1 CPU-minute.

Approach. I built a surrogate-accelerated pipeline: a Smolyak sparse-grid interpolant approximates the log-likelihood offline ($\sim 10^3$ PDE solves), then Metropolis-Hastings samples the surrogate at negligible cost.

- $< 5\%$ relative error in posterior mean and variance for a 50-D groundwater conductivity inversion.
- Clenshaw–Curtis nodes reduce interpolation error by $2\times$ over Legendre nodes for smooth likelihoods.
- Reusable Python module with FEniCS interface; tested on two published benchmarks.

ADDITIONAL PROJECTS

Finite-Element Solver for Incompressible Navier–Stokes

2019–2020

Senior honors thesis, State University

[FEM] [Fluid Dynamics] [C++] [deal.II]

Taylor–Hood ($\mathbf{P}_2/P_1$) mixed FEM for 2-D incompressible flow using deal.II; second-order velocity convergence verified via manufactured solutions. Lid-driven cavity flow at $\text{Re} = 400, 1000, 3200$ matches Ghia et al. (1982) to 4 significant figures.

Computer-Assisted Hopf Bifurcation Proof

Summer 2019

REU project, Another University

[Dynamical Systems] [Interval Arithmetic] [MATLAB/Intlab]

Used INTLAB to rigorously enclose Jacobian eigenvalues of a 3-species Lotka–Volterra system and certify a sign change in the real part, constituting a computer-assisted proof of Hopf bifurcation at $\mu^* = 0.7134 \pm 10^{-4}$.

TECHNICAL STACK

Domain	Tools & Methods
Numerical PDE	FEniCS, deal.II, PETSc, Firedrake, FDM, FEM, Spectral methods
HPC	C++17, MPI, OpenMP, SLURM, vectorised NumPy, JAX JIT
UQ / Stoch.	MLMC, Sparse Grids, Polynomial Chaos, MCMC, Kalman filters
Optimisation	SciPy, JAX (autodiff), L-BFGS, proximal methods, ADMM
ML	PyTorch, scikit-learn, Gaussian processes (GPyTorch)
Languages	Python, C/C++, MATLAB, Julia, R, Bash

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May 28, 2026

Prof. Recipient Name
Search Committee Chair
Department of Applied Mathematics
Target University or Company
City, State, ZIP

Re: Application for Postdoctoral Researcher in Numerical Analysis (Position #12345)

Dear Prof. Recipient Name,

I am writing to apply for the postdoctoral position in numerical analysis at Target University. I am completing my Ph.D. in Applied Mathematics at the University of Somewhere under the supervision of Prof. Jane Doe, with an expected graduation of May 2026. My research sits at the intersection of *numerical methods for stochastic PDEs*, *uncertainty quantification*, and *high-performance scientific computing* — areas directly aligned with your group's work on multilevel Monte Carlo methods for subsurface flow problems.

Dissertation research. My dissertation develops *adaptive* multilevel Monte Carlo (MLMC) estimators for elliptic stochastic PDEs $-\nabla \cdot (a(\mathbf{x}, \omega) \nabla u) = f$, where a is a log-normal random field. Standard MLMC requires pre-specified mesh hierarchies and sample counts, which is fragile when the coefficient is spatially rough. I designed an online algorithm that selects level parameters by minimising a cost-versus-variance trade-off estimated from a Gaussian process surrogate, and proved optimal $\mathcal{O}(\varepsilon^{-2} |\log \varepsilon|^2)$ complexity without *a priori* knowledge of coefficient regularity. Implemented in C++/MPI with FEniCS and PETSc, the method achieves a $3\times$ wall-clock speedup over standard MLMC on a 3-D Darcy flow benchmark with weak scaling to 512 cores. This work will appear in the *SIAM Journal on Scientific Computing* (2024).

Optimization and machine learning. During a research internship at National Lab, I proved sharp $\mathcal{O}(k^{-1/2})$ convergence rates for stochastic gradient descent under (L, σ) -smoothness — strictly weaker than the standard bounded-gradient assumption — and showed the rate is unimprovable by constructing matching lower bounds. This preprint is under revision for *NeurIPS 2024* and informs my long-term interest in theoretically grounded optimisation for large-scale inverse problems.

Why Target University. I am particularly drawn to Prof. Recipient's recent work on [specific paper or project]. The questions you raise about [specific technical aspect] connect directly to the algorithmic ideas I have developed, and I see clear potential for collaboration on [concrete joint direction]. I also look forward to contributing to graduate teaching; I have served as instructor of record for Calculus II and am keen to develop courses in numerical methods or scientific computing.

I have enclosed my curriculum vitae, project portfolio, and three letters of reference from Prof. Jane Doe, Dr. John Smith, and Prof. Alice Brown. I would welcome the opportunity to discuss my research at your convenience.

Sincerely,

Your Full Name

Ph.D. Candidate, Applied Mathematics
University of Somewhere
you@university.edu